

Executive Summary

The Objective:

To design and deploy a scalable, cloud-native **Security Information and Event Management (SIEM)** architecture that bridges the gap between decentralized endpoints and centralized governance. This project was developed to simulate the real-world regulatory requirements of **Loi 05-20** and **NIST-SP 800-53**, specifically focusing on **Continuous Monitoring** and **Log Integrity**.

The Architecture:

The core of the environment consists of a **Wazuh Manager** hosted on an **Azure Virtual Machine (Ubuntu-based)**. This centralized hub orchestrates security telemetry from diverse endpoints. This includes a local **Windows 10** host and a **Windows Subsystem for Linux (WSL)** instance. Communication is secured via encrypted agents and managed through refined **Azure Network Security Group (NSG)** rules to ensure the principle of “**Least Privilege**”.

Governance & Compliance Alignment:

Beyond technical deployment, this lab serves as a functional proof-of-concept for the following compliance controls:

- **Continuous Monitoring (CA-7 / ISO 27001 A.12.4.1):** Real time visibility into unauthorized access attempts and system anomalies.
- **File Integrity Monitoring (FIM):** Tracking unauthorized changes to critical system files to ensure system Integrity and Availability.

Monitoring Homelab - Wazuh SIEM for Log Analysis and Threat Detection

